

# NGUYEN SON TUNG

Hanoi, Vietnam · 0384886117

[tungns@soict.hust.edu.vn](mailto:tungns@soict.hust.edu.vn)

I am a dedicated researcher and educator with a strong background in Machine Learning, Deep Learning, and Natural Language Processing. I hold a Bachelor's degree in Computer Engineering (CPA 3.53/4.00) and a Master's degree in Data Science (CPA 3.72/4.00) from Hanoi University of Science and Technology (HUST). With five years of experience in both academic research and real-world applications, I have worked extensively on developing AI solutions while also engaging in teaching and mentoring students. Currently, I am a young faculty member at the School of Information & Communication Technology, HUST, where I contribute to both research and education in AI and data science.

## EXPERIENCE

06/2020 – 02/2021

**COLLABORATOR**, NCT LAB (HUST)

- Be a leader of a Smart Agriculture project that apply image processing technique and machine learning, deep learning to extract Vietnamese green mustard's phenotype (number of leaves, total leaf length and coverage area).
- I collect data, choose solution and build model to resolve the problem.
- I also support another team to deploy product in real life.

01/2020 – 06/2021

**MEMBER**, DATA SCIENCE LAB (HUST)

- Research and survey about some Continual Learning models, a new direction in machine learning.
- Make a project about Streaming Environment of Topic Model – a sub domain of NLP which is used to discover hidden topics in texts, and write a paper about this work (the paper was accepted in Neurocomputing journal).

09/2021

**TEACHING ASSISTANT**, VINBIGDATA.

- Assist in teaching the Fundamentals of Machine Learning course for the second batch of VinBigData AI Engineer program.

08/2022 – PRESENT

**LECTURER**, VIETAI.

- Assist in teaching the Deep Learning course, LLM course.
- Lecturer of Advances in Natural Language Processing course.

07/2021 – 06/2024

**AI ENGINEER**, BIG DATA DEPARTMENT, MOBIFONE IT CENTER

- Automatic Speech Recognition (ASR) project:

- Manage the data labeling team, monitor the labeling process, and prepare data for training Deep Learning models for speech-to-text conversion.
- Select, train, and evaluate the Wav2Vec2.0 model with telecommunication data.
- Train language models to improve the accuracy of the speech-to-text recognition system.
- Build a spelling correction model and add punctuation marks "." and "," for the output of speech-to-text system.
- Build a specific NER model to standardize the output of the speech-to-text system (standardize proper names, dates, and numbers, ...).
- Train a speaker diarization model for 2-speaker scenarios, applying it to the call center agent supervision task.
- Developing backend for speech-to-text website using FastAPI.
- **Real estate data analysis projects: Cus360 Dat Xanh and Mobiscare Houzz:**
  - Extract potential customers for real estate projects based on telecommunications data.
  - Select, extract essential information from the company's Big Data telecommunications system.
  - Choose, train, and evaluate several machine learning models for the purpose of extracting potential customers.
- **Enterprise Customer Service (ECM) project:**
  - Build the infrastructure for the entire system, select technology, create a list of system functionalities and operations.
  - Build Redis, Elasticsearch and database for the system.
  - Build the Minio Object Storage.
  - Integrate LDAP system and OnlyOffice platform into ECM system.
- **MobiFone chatbot project:**
  - Build the RAG chatbot system utilizing ChatGPT and semantic search model based on XLM-RoBERTa.
  - Transform the model to ONNX format to make the system run faster.
  - Build custom chatbot for each domain of the company using OpenAI API, Langchain, Elasticsearch and other vector databases.

**07/2024 – PRESENT**

**FACULTY TRAINEE**, HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY (HUST), VIETNAM.

- Working at School of Information & Communication Technology, HUST as a faculty trainee.

## EDUCATION

**08/2017 – 06/2021**

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Bachelor degree in Computer Engineering

CPA: 3.53/4.00

**08/2021 – 10/2023**

HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Master's Degree in Data Science

CPA: 3.72/4.00

## PUBLICATION

**Sharpness-Aware Minimization for Topic Models with High-Quality Document Representations.** Tung Nguyen, Tue Le, Hoang Tran Vuong, Quang Duc Nguyen, Duc Anh Nguyen, Linh Ngo Van, Sang Dinh, and Thien Huu Nguyen. In *The 2025 Annual Conference of the Nations of the Americas Chapter of the ACL*, 2025. <https://openreview.net/forum?id=JhieruHAG8> (Accept)

**GloCOM: A Short Text Neural Topic Model via Global Clustering Context.** Quang Duc Nguyen, Tung Nguyen, Duc Anh Nguyen, Linh Ngo Van, Sang Dinh, and Thien Huu Nguyen. In *The 2025 Annual Conference of the Nations of the Americas Chapter of the ACL*, 2025. <https://openreview.net/forum?id=3QThKopHsM> (Accept)

**Out-of-vocabulary handling and topic quality control strategies in streaming topic models,** Tung Nguyen, Tung Pham, Linh Ngo Van, Ha-Bang Ban, Khoat Than, *Neurocomputing*, Volume 614, 2025, 128757, ISSN 0925-2312, <https://doi.org/10.1016/j.neucom.2024.128757> ,  
( <https://www.sciencedirect.com/science/article/pii/S0925231224015285> )

**NeuroMax: Enhancing Neural Topic Modeling via Maximizing Mutual Information and Group Topic Regularization.** Duy-Tung Pham, Thien Trang Nguyen Vu, Tung Nguyen, Linh Van Ngo, Duc Anh Nguyen, and Thien Huu Nguyen. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 7758–7772, Miami, Florida, USA. Association for Computational Linguistics. <https://aclanthology.org/2024.findings-emnlp.457.pdf>

**Balancing stability and plasticity when learning topic models from short and noisy text streams,** Tung Nguyen, Trung Mai, Nam Nguyen, Linh Ngo Van, Khoat Than, *Neurocomputing*, Volume 505, 2022, Pages 30-43, ISSN 0925-2312, <https://doi.org/10.1016/j.neucom.2022.07.019> ,  
(<https://www.sciencedirect.com/science/article/pii/S0925231222008785> )

## LANGUAGE

- English: Can read, write, speak, understand and peer review.